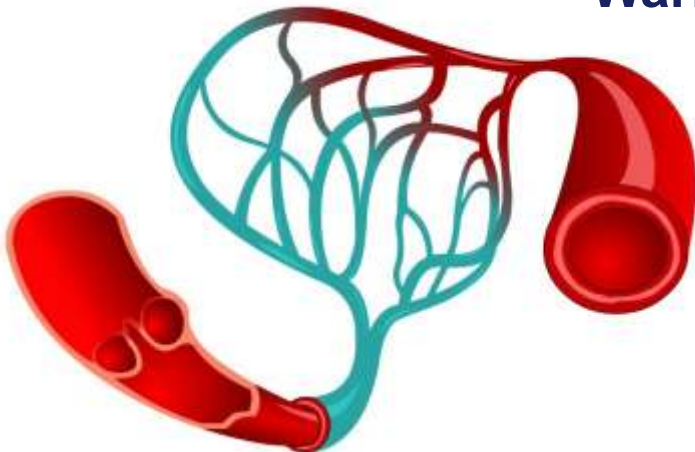


Vasodilators



Warisara Parichatikanond, Ph.D., Dr.Scient.Med.

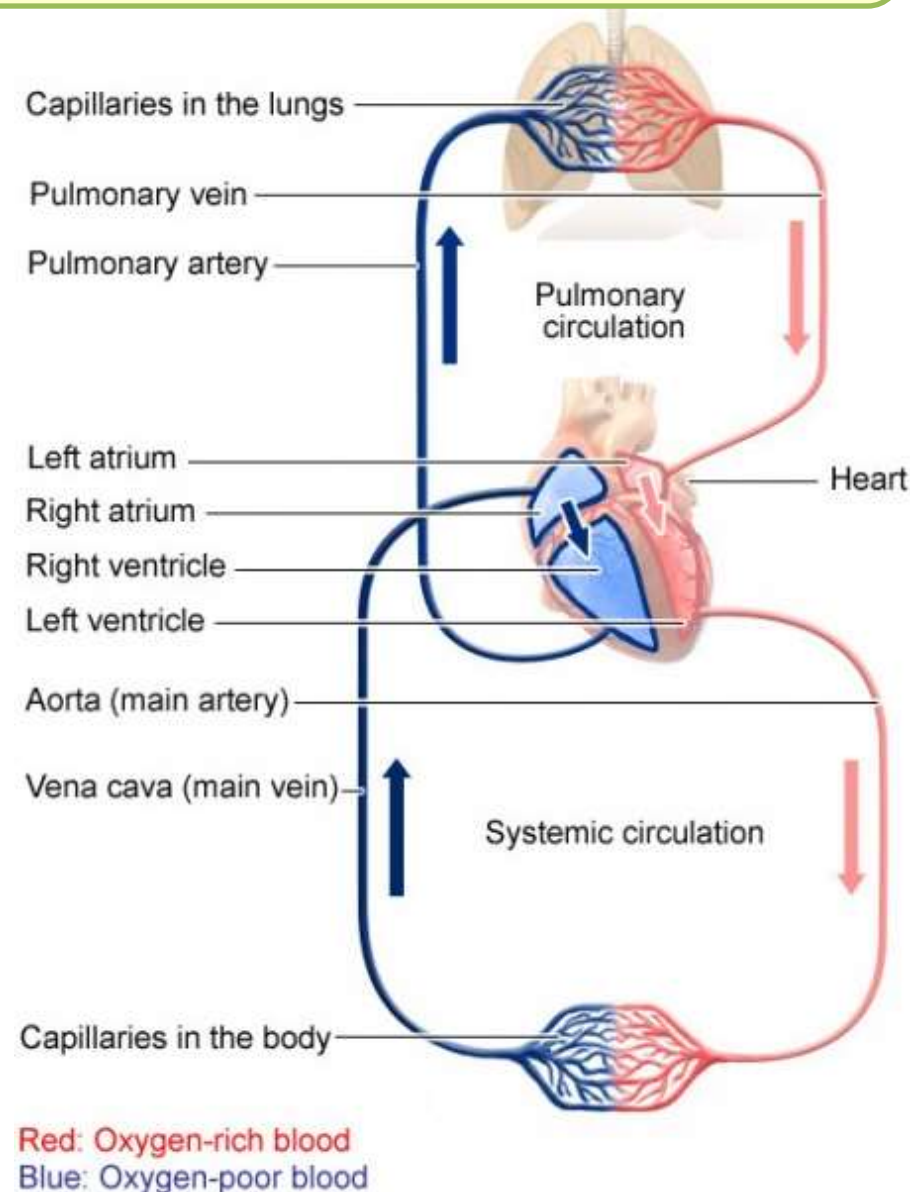
Department of Pharmacology

Faculty of Pharmacy, Mahidol University

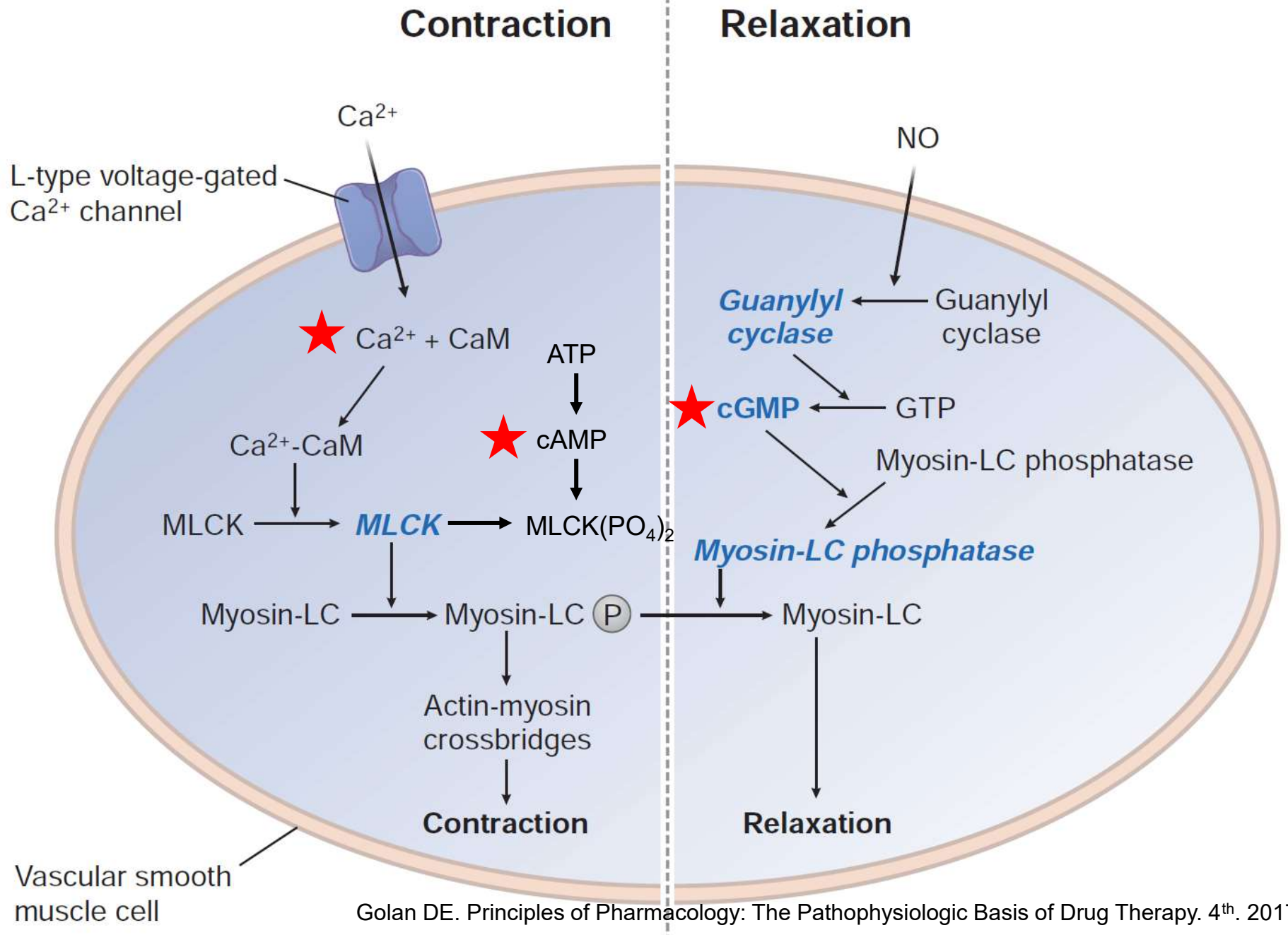
E-mail: warisara.par@mahidol.edu

Blood vessels

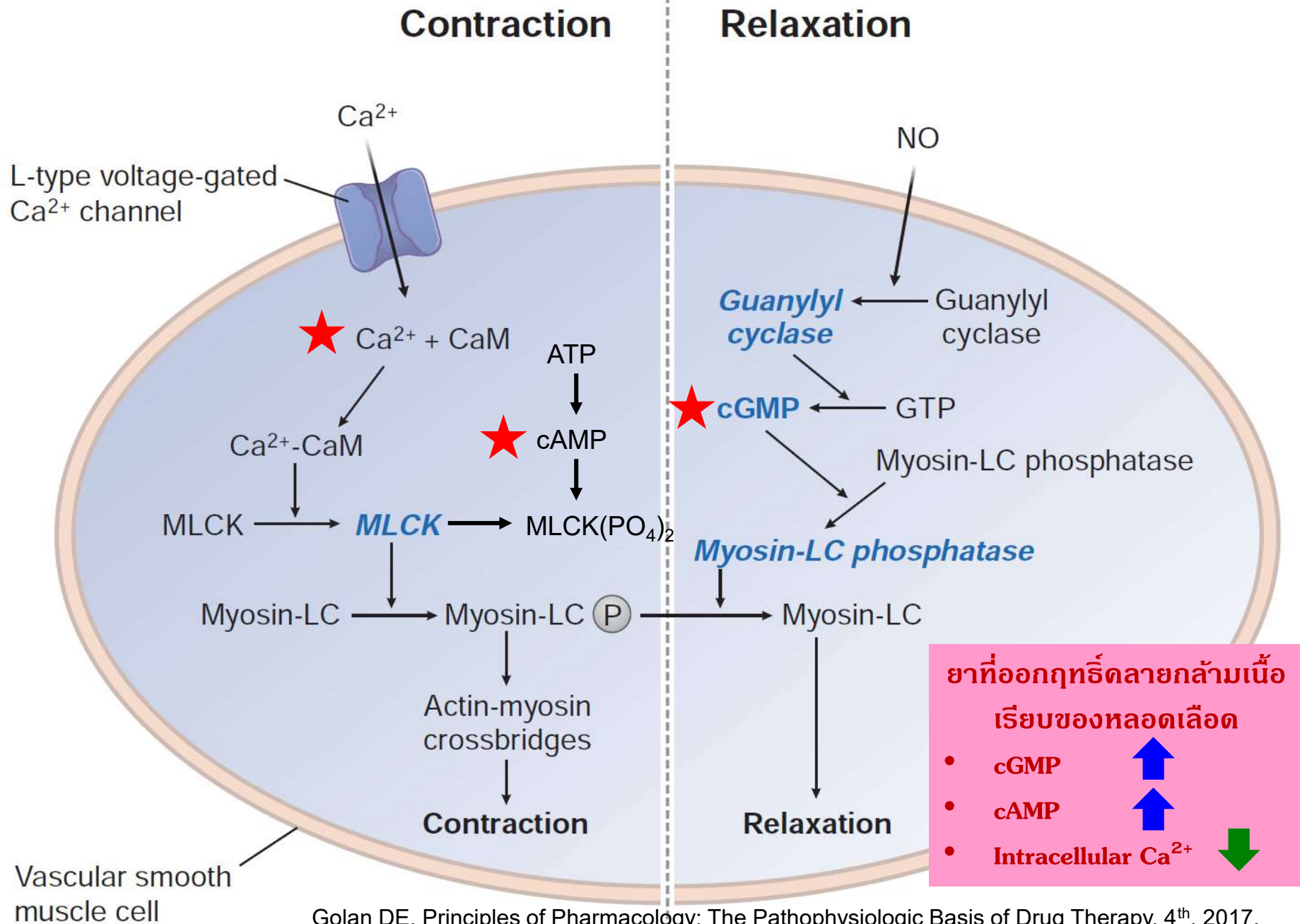
- **Blood vessels:** Transport blood throughout body in circulatory system
- **Arteries:** Carry blood away from the heart (e.g. aorta, artery, arteriole)
- **Capillaries:** Exchange of water and chemicals between blood and tissues
- **Veins:** Carry blood from the capillaries back toward the heart (e.g. vena cava, vein, venule)



Vascular smooth muscle cell

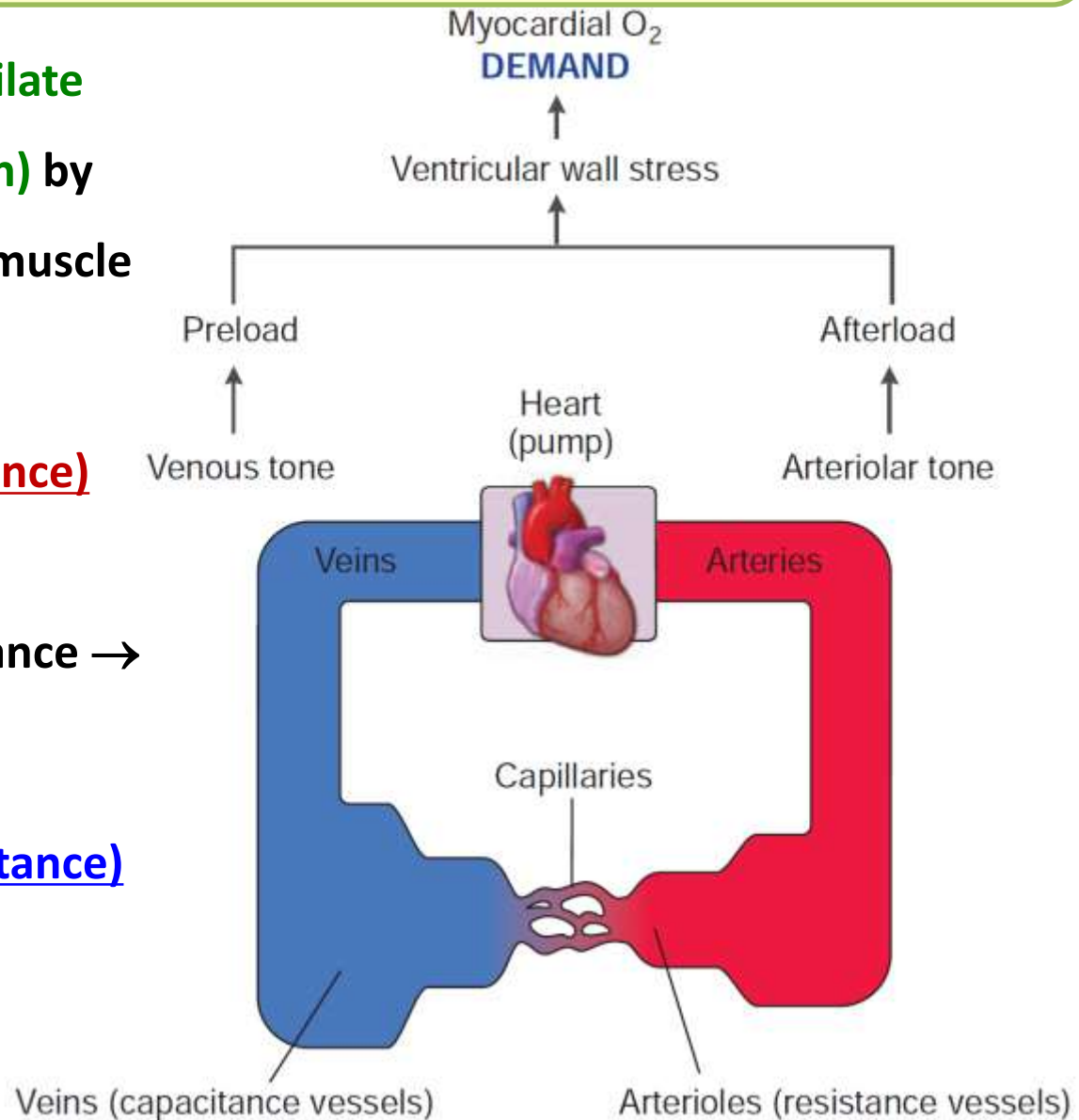


Vascular smooth muscle cell



Vasodilators

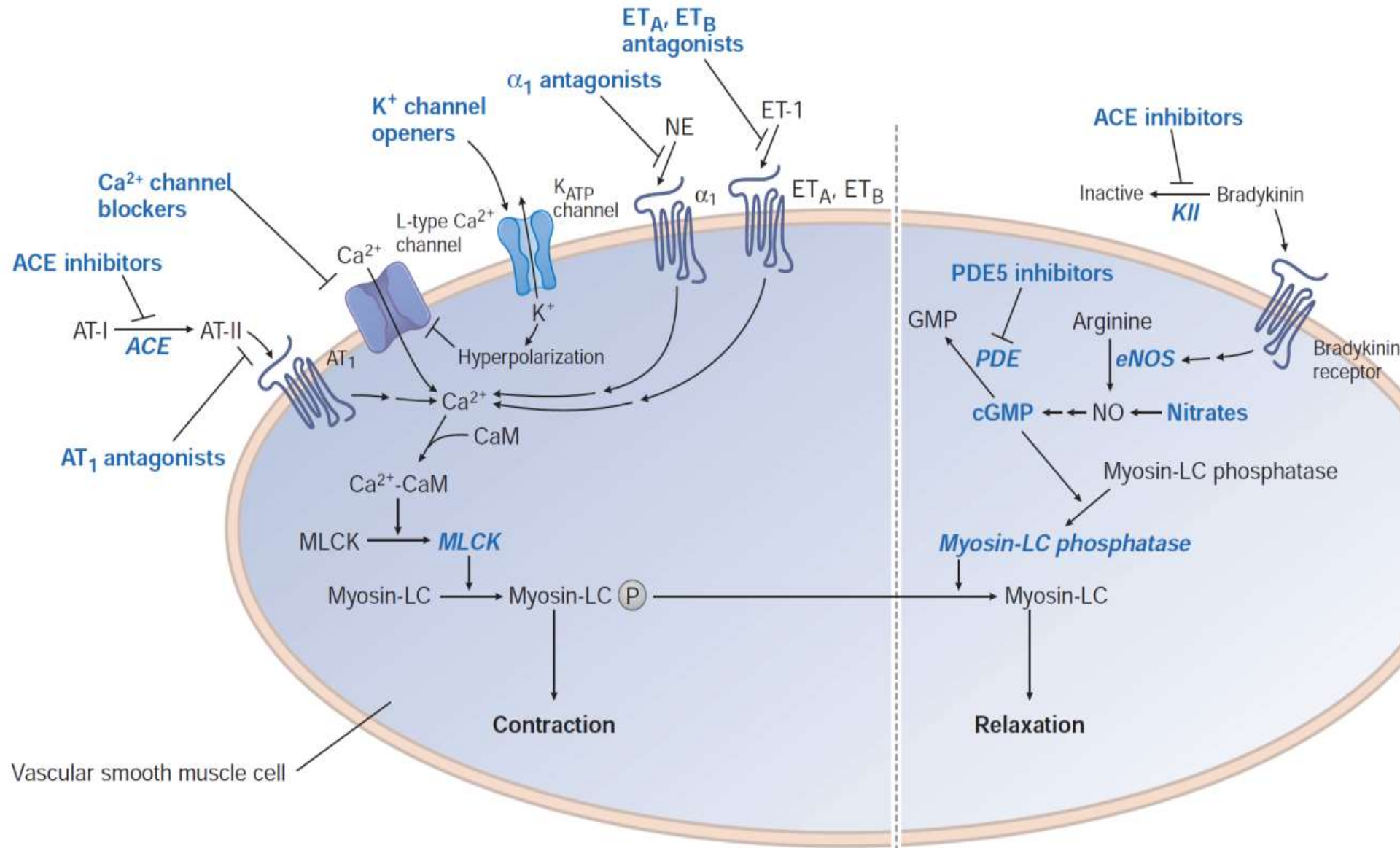
- **Vasodilators:** Drugs that **dilate blood vessels (vasodilation)** by relaxing vascular smooth muscle cells
- Dilation of arterial (resistance) **vessels** → ↓ **Afterload** → ↓ systemic vascular resistance → ↓ arterial blood pressure
- Dilation of venous (capacitance) **vessels** → ↓ **Preload** → ↓ venous blood pressure



Vasodilators: Classification

- α_1 -AR antagonists: Prazosin, doxazosin, terazosin
- Ca^{2+} channel blockers (CCBs): Nifedipine, diltiazem, verapamil
- RAAS blockers: ACEIs (-pril), ARBs (-sartan), DRI (aliskiren)
- Hydralazine
- K^+ channel openers: Minoxidil, pinacidil, nicorandil
- Sodium nitroprusside
- Organic nitrates: Nitroglycerin (NTG), isosorbide dinitrate (ISDN), isosorbide mononitrate (ISMN)
- PDE-5 inhibitors: Sildenafil, vardenafil, tadalafil
- Endothelin receptor antagonists: Bosentan, ambrisentan, macitentan
- Soluble guanylyl cyclase (sGC) stimulators: Riociguat, vericiguat
- Neprilysin inhibitors: Sacubitril

Sites of action of vasodilators



Arterial vasodilator

- **Hydralazine**
- **Minoxidil**

Hydralazine: Pharmacological effects & PK

- **Mechanism of action:**
 - Direct relaxation of arteriolar smooth muscle
 - ↑ NO release of endothelial cells (Arterioles) → ↑ cGMP
- **Pharmacological effects:**
 - Vasodilation → ↓ TPR → ↓ BP
 - Dilation of arterioles → Orthostatic hypotension is less**
- **Pharmacokinetics:**
 - Well absorbed, low bioavailability (first-pass hepatic metabolism)
 - 16% (bioavailability) in Fast acetylators (Asians)
 - 35% (bioavailability) in Slow acetylators (Blacks & Caucasians)
 - Hydralazine is N-acetylated by N-acetyltransferase 2 (NAT2)
 - Rate of metabolism depends on slow/fast acetylator

Hydralazine: Adverse effects

- Reflex tachycardia (due to baroreceptor reflex)
 - **Prevention:** Co-administer with **β -blocker**
- Systemic lupus erythematosus (SLE)-like syndrome:
 - Unknown mechanism
 - Occur after **long-term use (> 6 month)**
 - **SLE-like syndrome:** Sore throat, fever, rash, muscle and joint aches

Drug-induced lupus by hydralazine



Hydralazine: Therapeutic uses

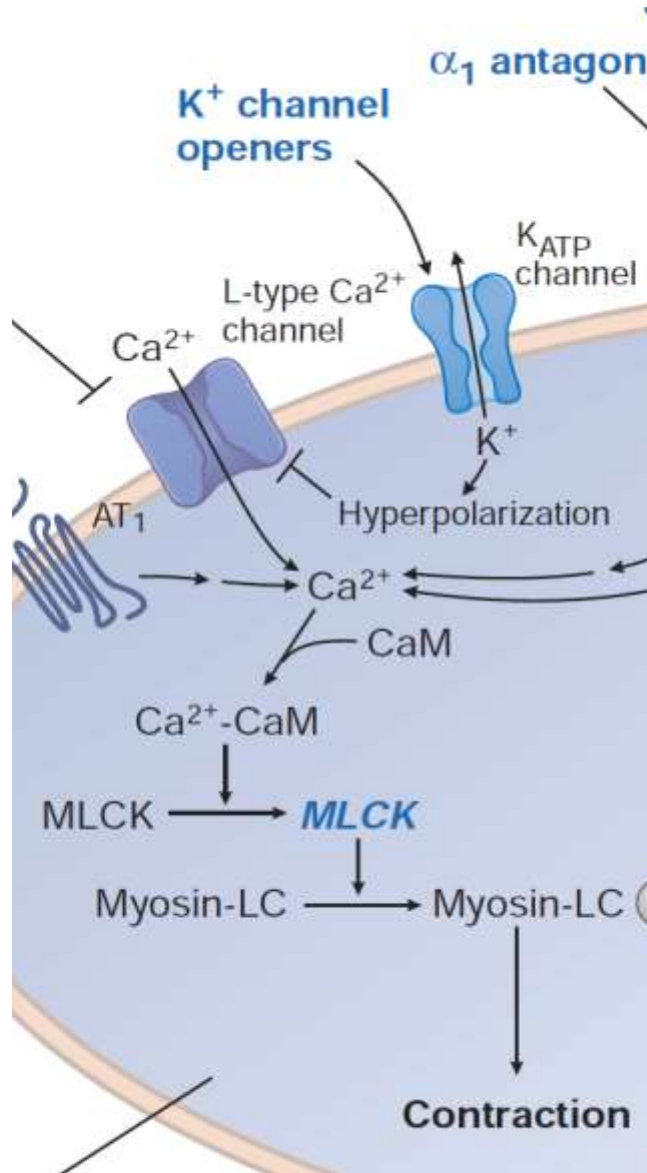
- **Hypertension:** For hypertensive emergencies in pregnancy
 - Hydralazine is **not used as a primary drug** for hypertension
- **Heart failure (HF):** Hydralazine + Isosorbide dinitrate (H + ISDN)

ESC Guideline for HF (H + ISDN):

- Consider in black patients (**African-American**) with **LVEF $\leq$ 35%**
- Consider in HFrEF patients who **are intolerant of ACEIs and ARBs** due to renal dysfunction or hyperkalemia

Drugs	Indication	Dosage Form	บัญชียาหลัก 2568
Hydralazine (Apresoline®)	Hypertensive emergencies in pregnancy	Tab 25 mg Inj. 20mg/5mL	ก

Minoxidil: Pharmacological effects & PK



Mechanism of action:

- Minoxidil metabolized by **sulfotransferase** to **minoxidil N-O sulfate (active)**
- Minoxidil N-O sulfate causes vasodilation by **activating ATP-modulated K^+ channel**

K^+ efflux $\rightarrow$ Membrane hyperpolarize (no depolarization) $\rightarrow$ Ca^{2+} channels closed $\rightarrow$ Inhibit contraction

Pharmacological effects:

- **Arteriolar vasodilation** $\rightarrow$ $\downarrow$ TPR $\rightarrow$ $\downarrow$ BP
- $\uparrow$ blood flow to **skin****, skeletal muscle, GI tract and heart

Minoxidil: Adverse effects

- **Salt and water retention:**
 - Prevention: Co-administer with **diuretics**
- **Cardiovascular effects:**
 - Vasodilation → baroreceptor reflex (sympathetic reflex) -> ↑ contractility & tachycardia
 - **Prevention:** Co-administer with **β-blocker**
- **Hypertrichosis (hirsutism):** Long-term use (↑ cutaneous blood flow)
 - Hair growth on the face, back, arms and legs

Hypertrichosis in a young female using topical minoxidil 5%



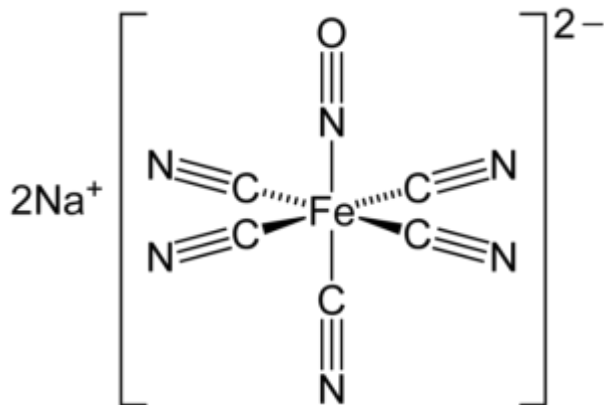
Minoxidil: Therapeutic uses

- **Severe hypertension** that responds poorly to other anti-hypertensives
(not for female due to hair growth)
- **Androgenetic alopecia (Baldness)** (Topical, Rogaine®)
 - Dosage: Adult: 1 ml of solution applied to affected area bid.
 - Long-term therapy: at least 4 months

Drugs	Indication	Dosage Form	บัญชียาหลัก 2568
Minoxidil (Loniten®)	Hypertension	Tab 5 mg	NED
Minoxidil (Rogaine®, Reten®)	Androgenetic alopecia	Topical 2, 5%	NED

Arterial & venous vasodilator (Mixed vasodilator)

- Sodium nitroprusside**



A nitrate compound consists of a nitrosyl (NO) group, 5 cyanide (CN) groups and iron

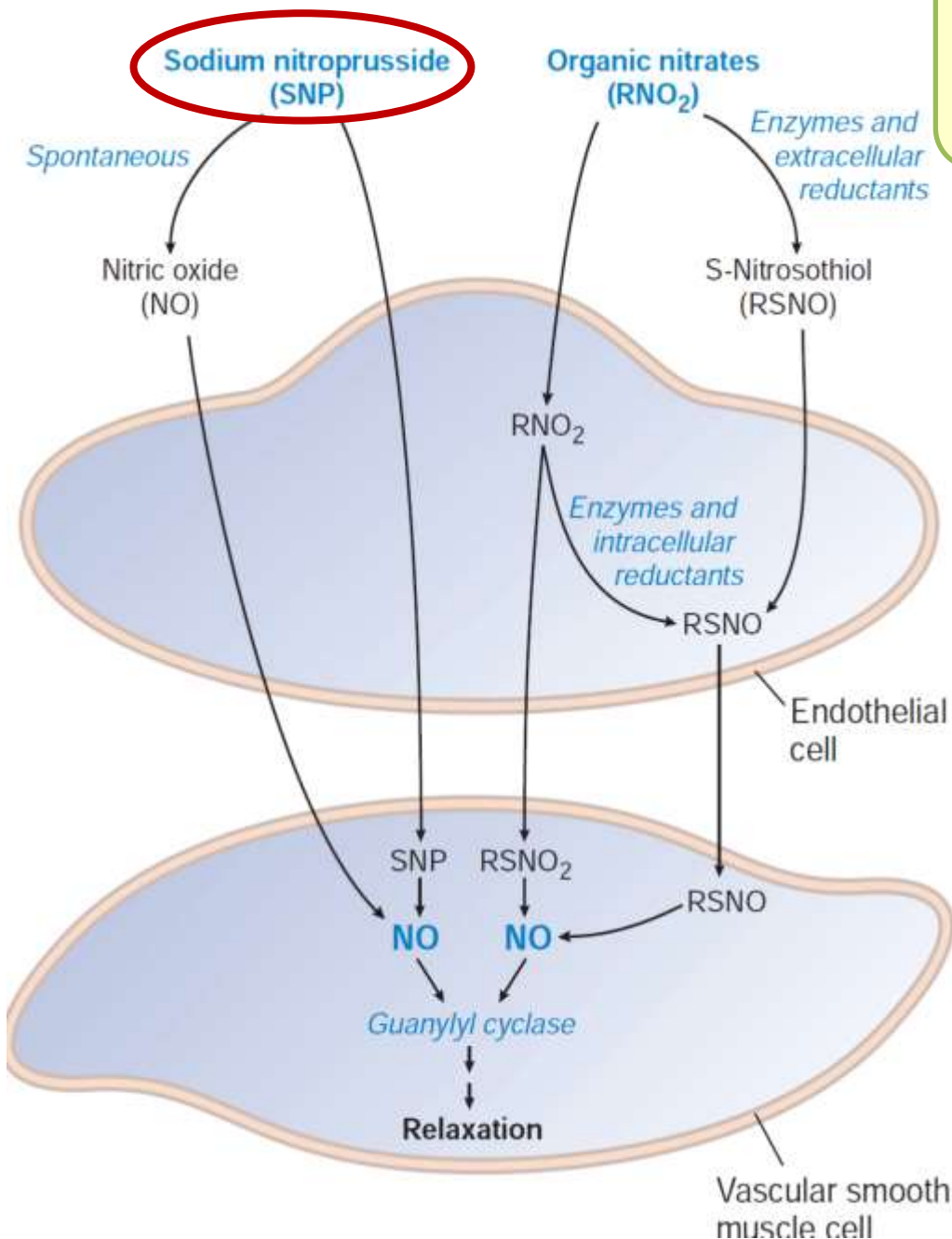
Nitroprusside: Pharmacological effects

Mechanism of actions:

- **Nitroprusside** is a potent vasodilator. Release NO spontaneously **without enzymatic demand**
- NO activates **guanylyl cyclase**, leading to the formation of cGMP and vasodilation

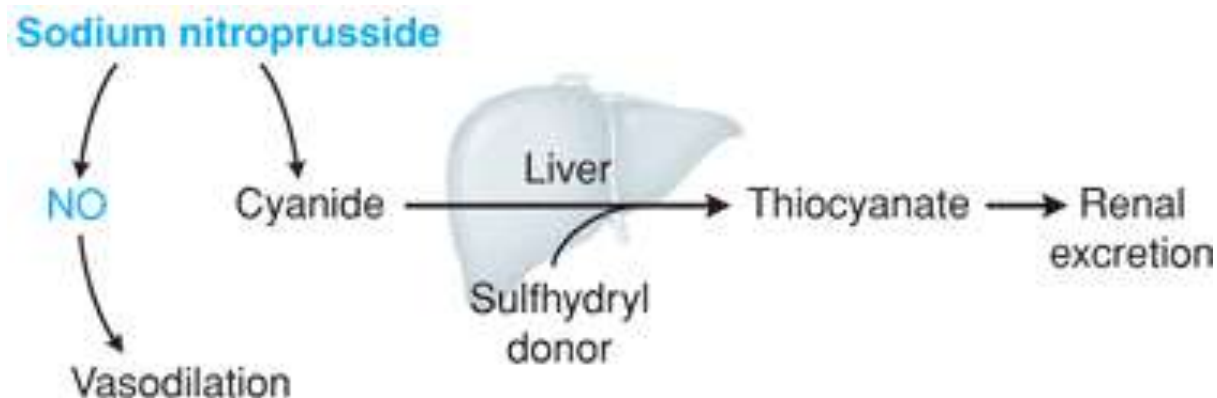
Pharmacological effects:

- Nitroprusside dilates **both arterioles and venules** -> venous pooling -> ↓ arterial resistance



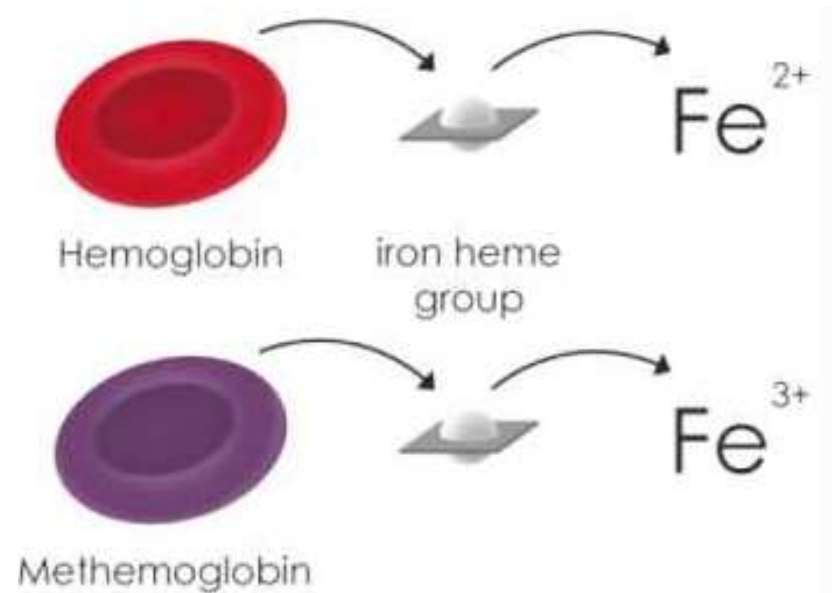
Nitroprusside: Pharmacokinetics

- Nitroprusside: is an **unstable molecule** -> decomposes in **strongly alkaline conditions and when exposed to light** (freshly prepared, check appearance**)
- Route of administration: **Continuous IV infusion**
 - Onset of action: 30 sec, Peak: 2 min
 - When infusion is stopped, effect disappears within 3 min
- Nitroprusside is metabolized to **NO and cyanide**
- **Cyanide toxicity*** can result upon **prolonged use** or **renal /liver disease**



Nitroprusside: Adverse effects

- Excessive vasodilation and hypotension
- Cyanide toxicity: Metabolic acidosis, arrhythmia, severe hypotension
 - Cause: Prolonged use, fast infusion rate, renal dysfunction
 - Antidotes: Hydroxocobalamin (vitamin B₁₂)
- Methemoglobinemia
 - Hemoglobin (Fe^{2+} ; ferrous) → Methemoglobin (Fe^{3+} ; ferric)



Nitroprusside: Therapeutic uses

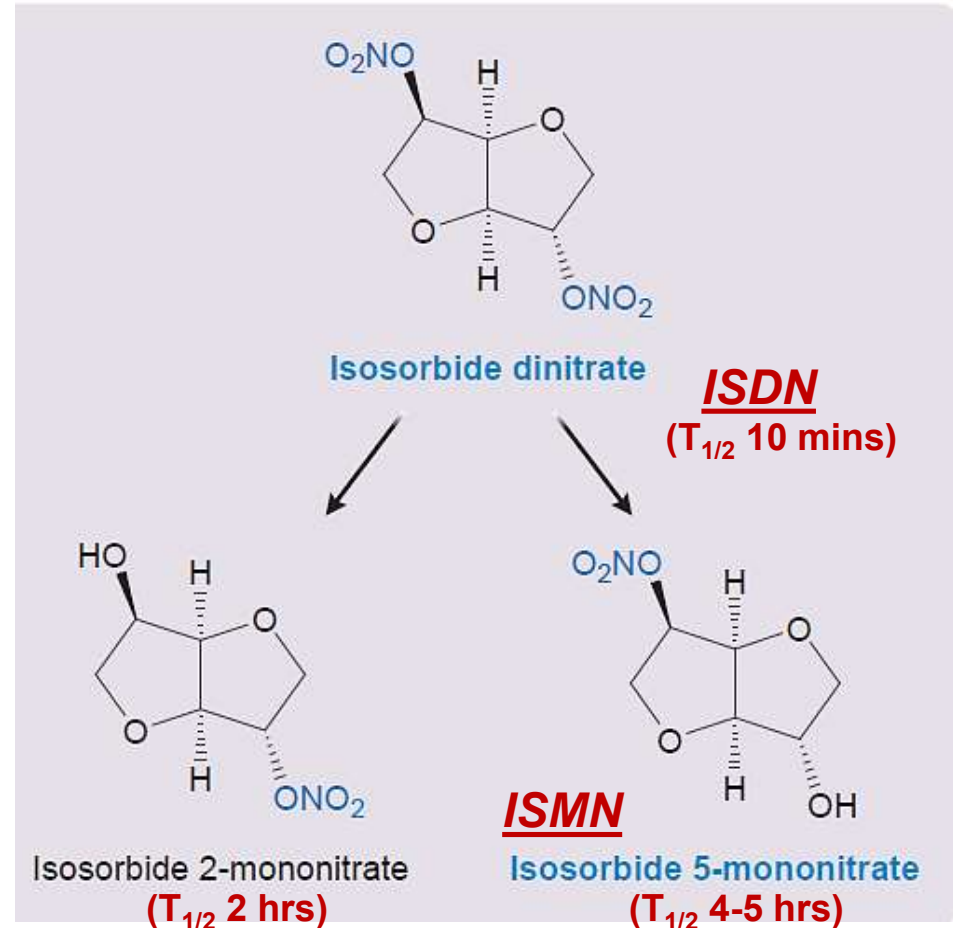
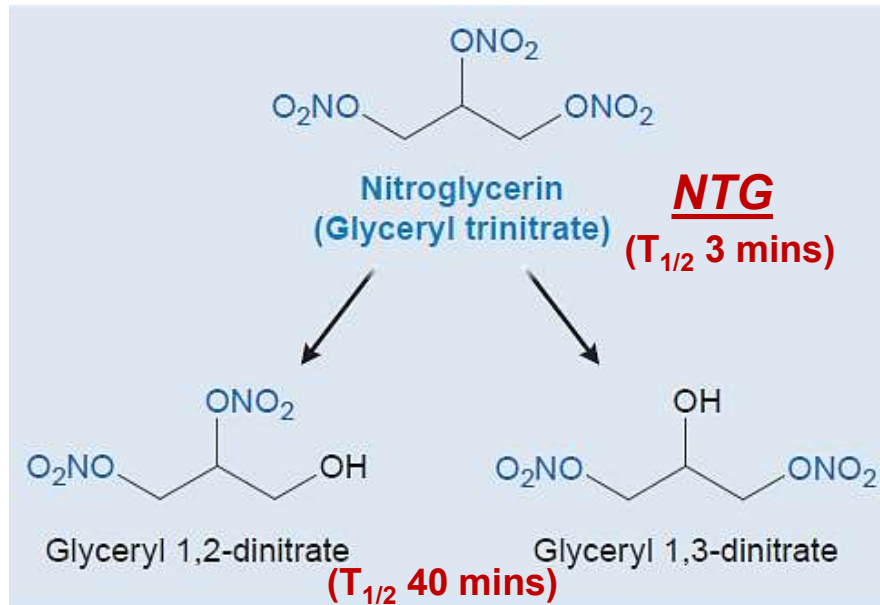
- Hypertensive emergencies, Except for pregnancy
 - to control hypotension during surgery
- Nitroprusside is useful only for **short-term treatment** (max, up to 72 hr)
- **Warning:** Giving excessive doses (>500 mcg/kg), too fast ($>2-5$ mcg/kg/min), or maximum infusion rate (10 mcg/kg/min for >10 min) can cause **cyanide toxicity**

Drugs	Indication	Dosage Form	บัญชียาหลัก 2568
Sodium nitroprusside	Hypertensive emergencies, Except for pregnancy	Inj 50 mg	ก

Organic nitrates

- **Nitroglycerin (NTG)**
- **Isosorbide dinitrate (ISDN)**
- **Isosorbide 5-mononitrate (ISMN)**

Organic nitrates: Structure



NTG and ISDN are biologically active nitrates that are metabolized into active molecules with **longer half-lives** than their parent compounds.

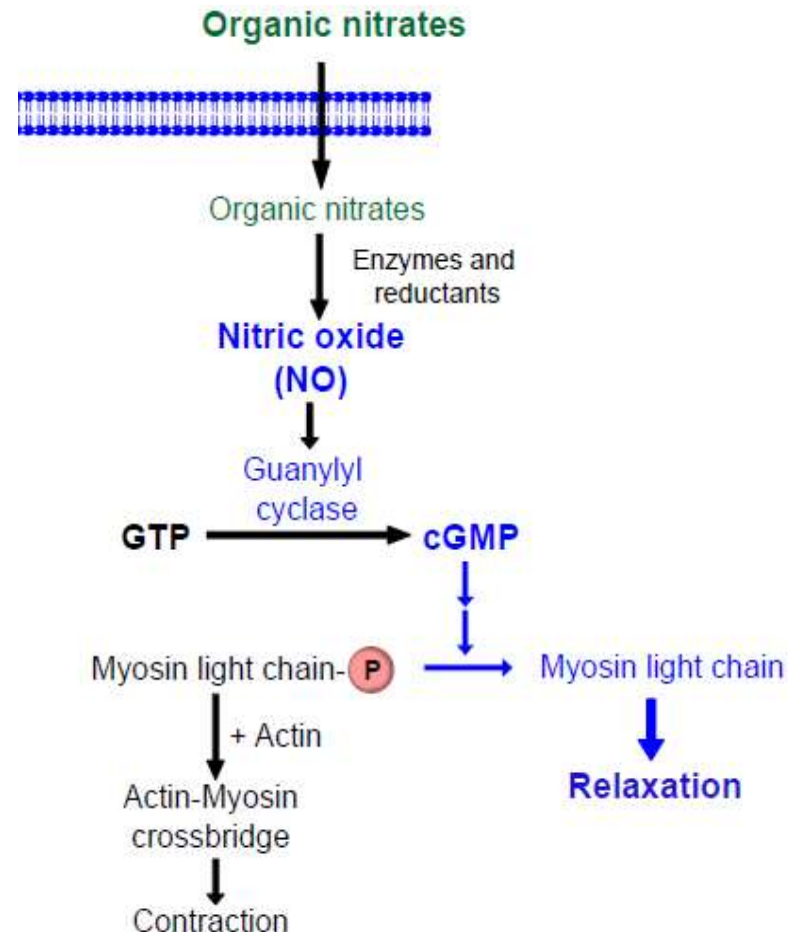
Organic nitrates: Pharmacological effects

Mechanism:

- Nitro group is reduced to form nitric oxide (NO) by specific enzymes and reductants (e.g., thiols)
- NO **activates guanylyl cyclase** → **↑ cGMP** → vascular smooth muscle **relaxation**
- NO also functions as an **antiplatelet agent**

Pharmacological effect:

- **Vasodilation**
 - Major: **Dilate of vein****:
↑ venous capacitance, ↓ **preload**
 - Minor: Dilate of artery (high dose):
↓ TPR → ↓ afterload
- ↓ **Myocardial oxygen demand**



Organic nitrates: Pharmacokinetics

- **NTG & ISDN:**

- Hepatic first-pass metabolism → Very low oral bioavailability
- **Avoiding first-pass metabolism: Sublingual & Transdermal**

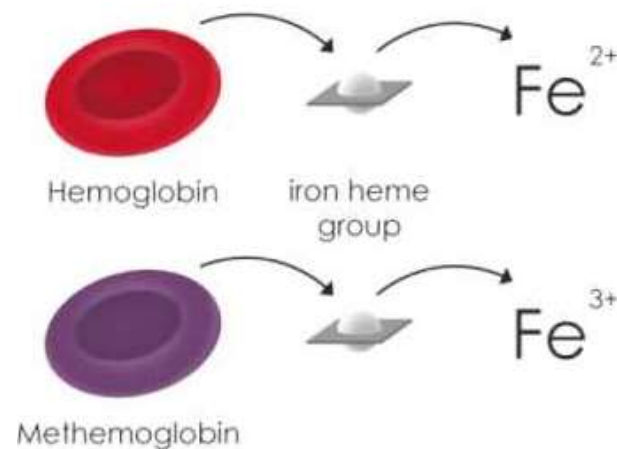
- **ISMN:**

- No first-pass metabolism → F 100% after oral administration

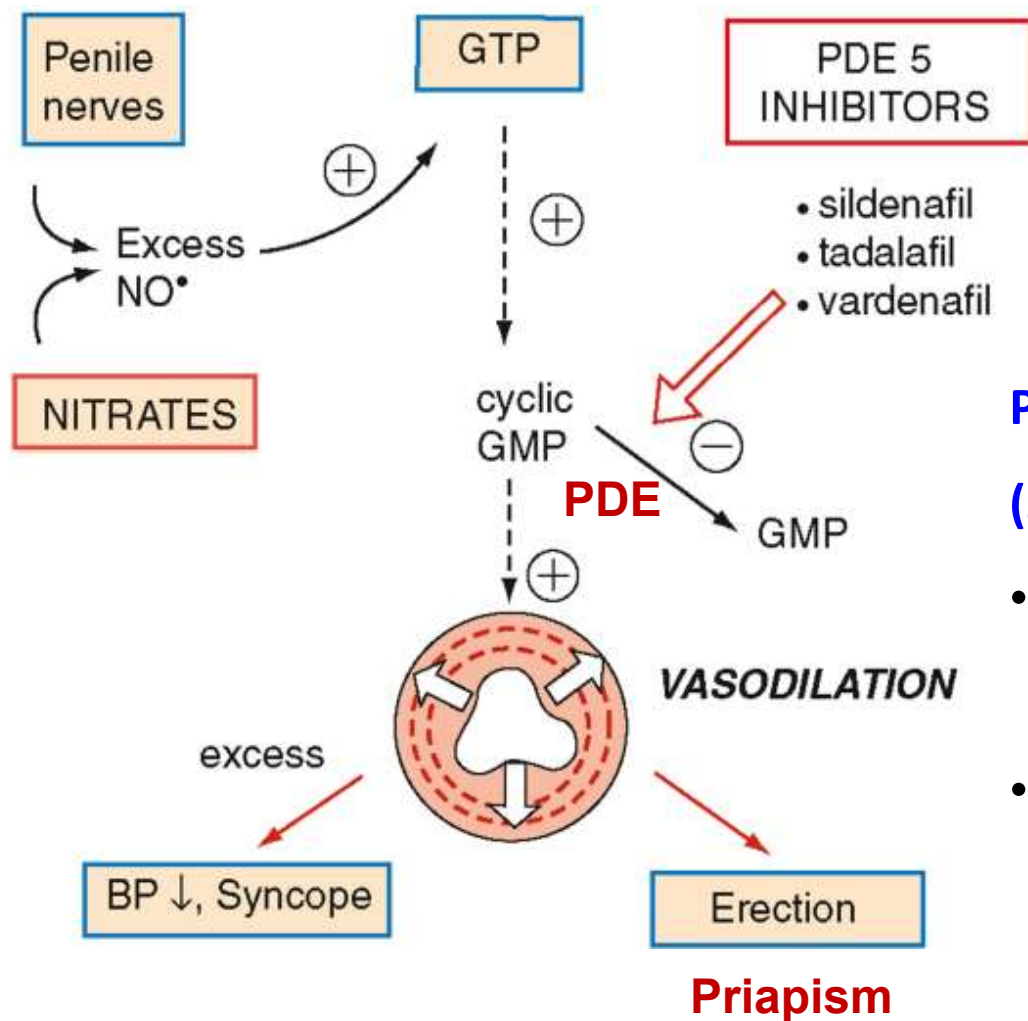
Organic Nitrate	NTG	ISDN	5-ISMN
Half-life (min)	3	10	280
Plasma clearance (L/min)	50	4	0.1
Volume of distribution (L/kg)	3	4	0.6
Oral bioavailability (%)	<1	20	100

Organic nitrates: Adverse effects

- **Orthostatic hypotension, syncope** (due to ↑ venous capacitance)
- **Reflex tachycardia** (baroreceptor reflex)
- **Most common:** Headache, dizziness
- **Occasional:** flushing, nausea, vomiting
- **Contact dermatitis:** 40% of transdermal patch
- **Methemoglobinemia** (rarely, but serious)
 - Antidote: **Methylene blue, hydroxocobalamin (vitamin B₁₂)**
- **Nitrate tolerance**
 - Continuous or frequent exposure
 - Solution: **Nitrate-free periods** of at least 8 hrs***
 - Tablet → TID, night- free
 - Transdermal patch (24-hr): Remove patch during the night



Organic nitrates: Serious drug interaction



PDE-5 inhibitors

(sildenafil, tadalafil, vardenafil)

- Cause life-threatening hypotension
- Use with nitrates in any form is **contraindicated !!**

Organic nitrates: Therapeutic uses

- Angina pectoris, heart failure, acute MI

Drugs	Indication	Dosage Form	บัญชียาหลัก 2568
Glyceryl trinitrate (NTG, GTN)	Angina pectoris	Injection 10, 50 mg/10ml Transdermal patch Aerosol	ก NED NED
Isosorbide dinitrate (ISDN, Isordil®)	Angina pectoris	Tablet 10, 20, 30 mg Tablet 5 mg (SL)	ก ก
Isosorbide 5-mononitrate (ISMN, Ismo®)	Angina pectoris	Tablet 20, 60 mg Capsule (SR) 20 & 60 mg	ก NED
Pentaerythritol tetranitrate (PETN)	Angina pectoris	Tablet 10 mg	NED

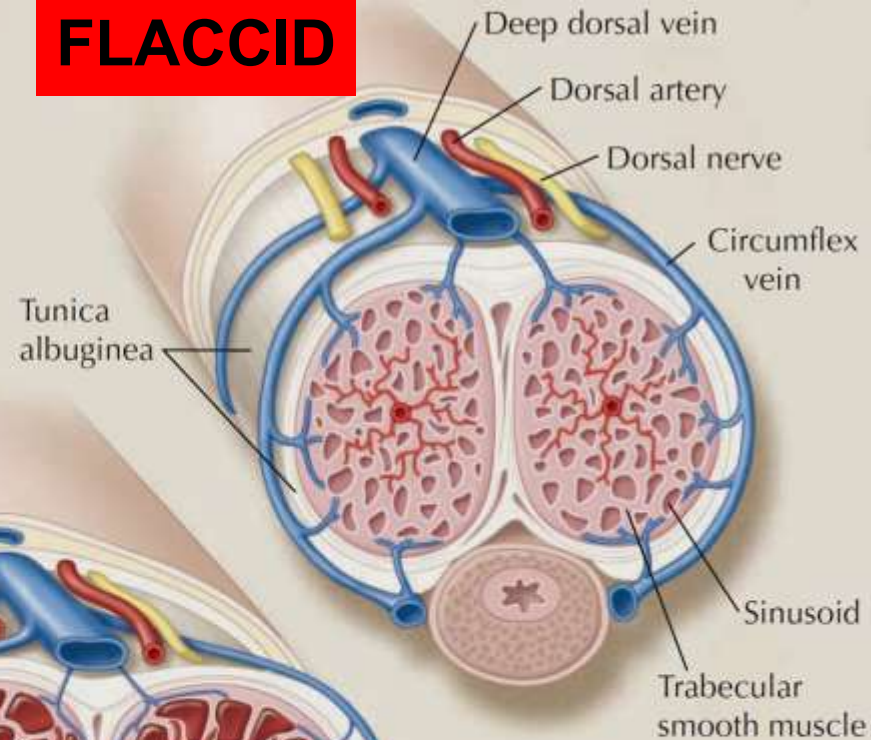
SL; Sublingual, SR; Slow release, NED; non-essential drugs

Phosphodiesterase 5 (PDE-5) inhibitors

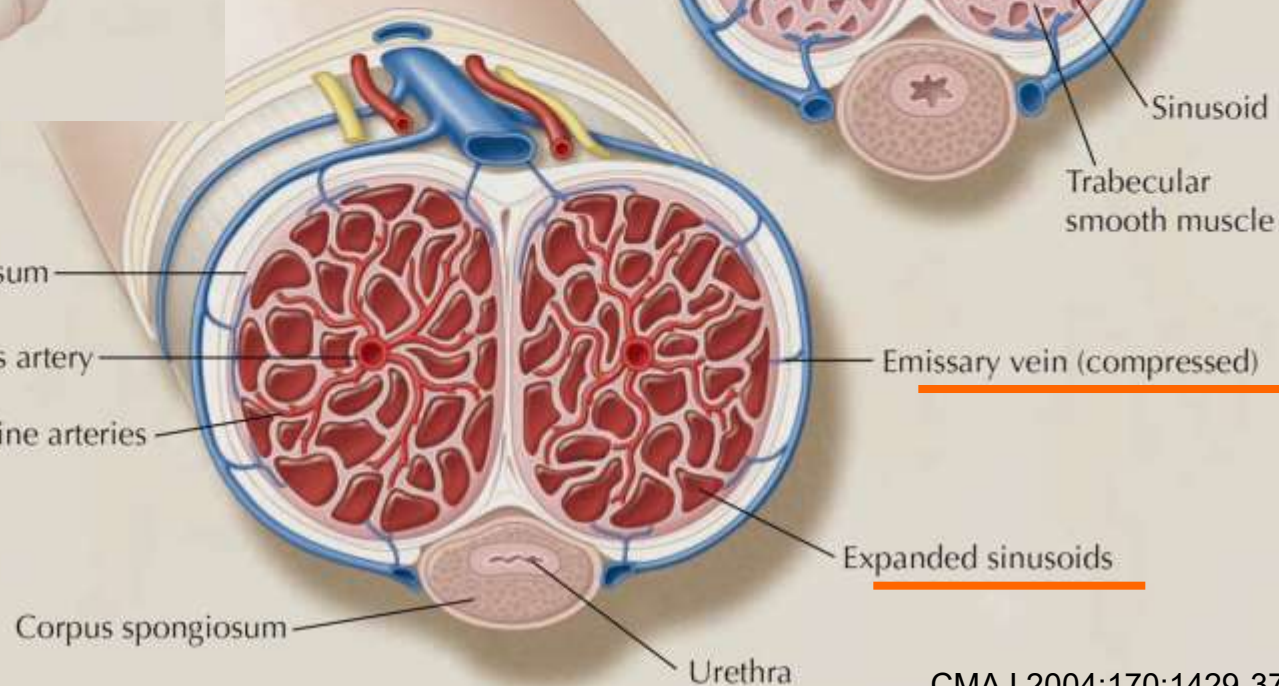
- **Silden**afil
- **Avan**afil
- **Tadal**afil
- **Varden**afil

Physiology of penile erection

FLACCID



ERECT



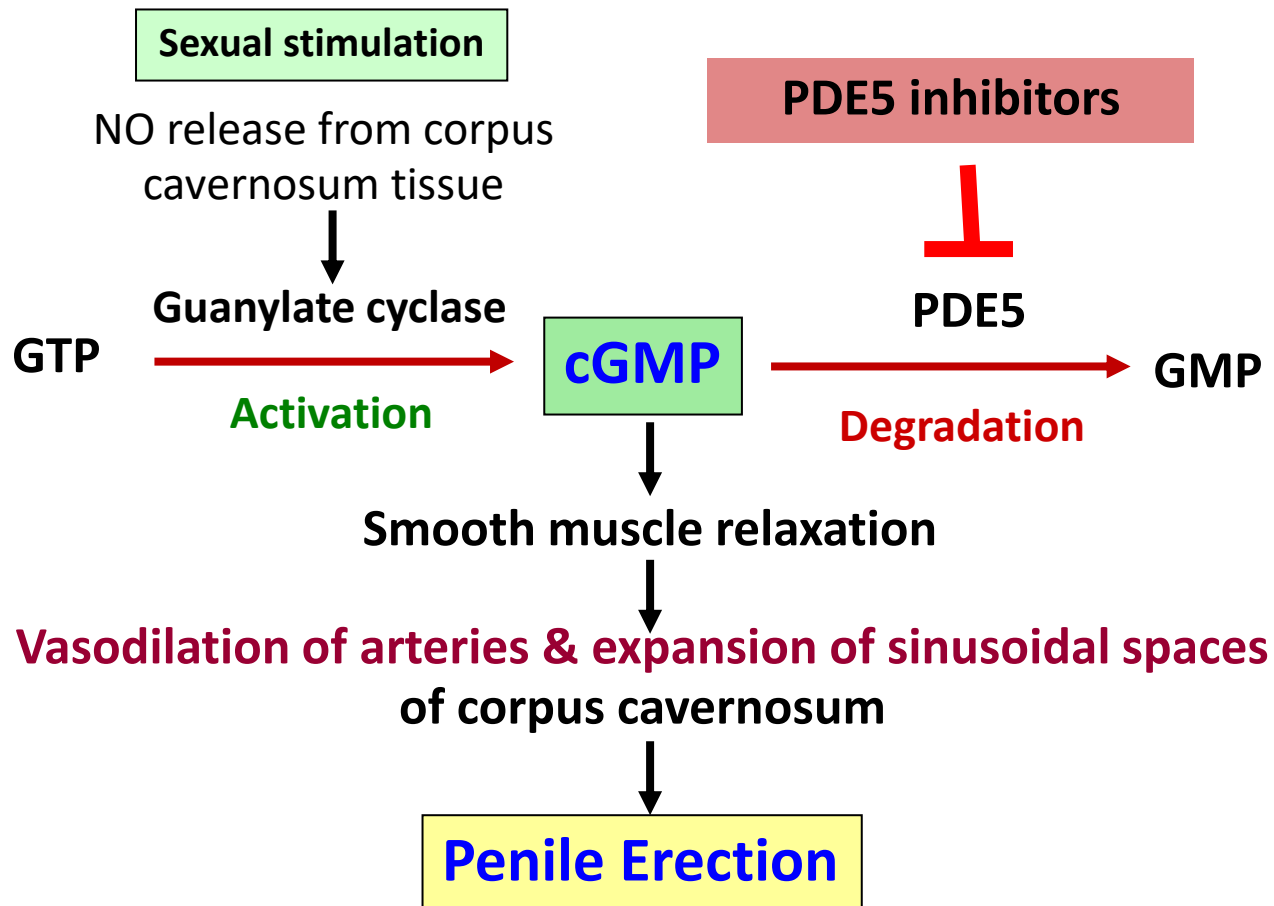
2 of corpus cavernosum &
1 of corpus spongiosum

Erection:

- Expanded sinusoids
- ↑ Arterial blood flow
- Compressed vein

PDE-5 Inhibitors: Pharmacological effects

Mechanism: Selective inhibit phosphodiesterase 5 (PDE5) enzyme activity, which is responsible for degradation of cGMP at the penis



PDE-5 Inhibitors: Pharmacokinetics

PDE5 Inhibitors	Avanafil (Stendra®)	Sildenafil (Viagra®)	Tadalafil (Cialis®)	Vardenafil (Levitra®)
Dosage (for ED)	- Start: 100 mg - Max: 200 mg/day	- Start: 50 mg - Max: 100 mg/day	- Start: 10 mg - Max: 20 mg/day	- Start: 10 mg - Max: 20 mg/day
Onset	15-30 min	30-60 min	45 min	30 min
Duration	4-8 hr	4-8 hr	Up to 36 hr	4-8 hr
Efficacy	>65%	>65%	>65%	>65%
Food & alcohol interaction	No	- Interact with food (take while fasting) - No alcohol interaction	No	- Interact with food (take while fasting) - No alcohol interaction

Advantages

- Oral administration
- Effective
- Good tolerability

Disadvantages

- Delayed onset of action
- Contraindication with nitrates
- Restriction of frequency of use (Not more than once per day)
- Adverse effects
- Drug interaction (esp, CYP3A4)

Differences in PDE subtype selectivity

- **PDE11:**
 - Found in testes and prostate
 - However, PDE5 inhibitors have no effect on spermatogenesis and testicular function
- **PDE6:**
 - Found in retina (PDE6 controls visual transduction)
 - Inhibits retinal PDE6 activity cause **visual disturbances: Blurred vision, flashing lights, change in color perception (e.g, blue, green)**
 - **Incidence: Sildenafil ~ Vardenafil > Avanafil > Tadalafil**

PDE-5 Inhibitors: Adverse effects

- **Hypotension**
- **Vision loss** (rarely)
- **Hearing loss** (rarely)
- **Heart attack, stroke, irregular heart beats:** in patient with heart disease (rarely)
- **Priapism:** A persistent and usually painful long-lasting erection (more than 4 hr) (rarely)

PDE-5 Inhibitors: Therapeutic uses

- Erectile dysfunction (ED)
- Benign prostatic hyperplasia (BPH): Tadalafil only
- Pulmonary arterial hypertension (PAH)

Drugs	Indication	Dosage Form	บัญชียาหลัก 2568
Sildenafil (Viagra®)	ED, PAH	Tab 25, 50, 100 mg	ง
Avanafil (Strenda®)	ED	Tab 50, 100, 200 mg	NED
Tadalafil (Cialis®)	ED	Tab 2.5, 5, 10, 20 mg	NED
Vardenafil (Levitra®, Staxyn®)	ED	Tab 2.5, 5, 10, 20 mg	NED

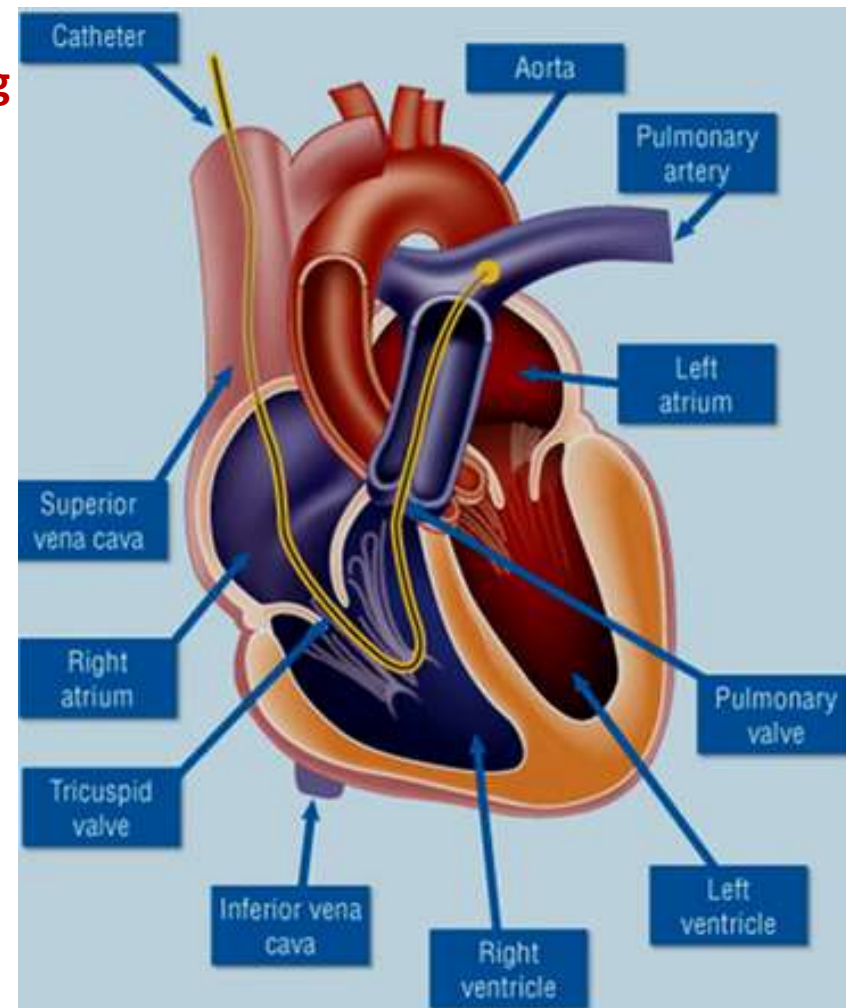
Endothelin receptor antagonists (ERAs)

- **Bo**sentan
- **Ambri**sentan
- **Macit**entan

Pulmonary Arterial Hypertension (PAH)

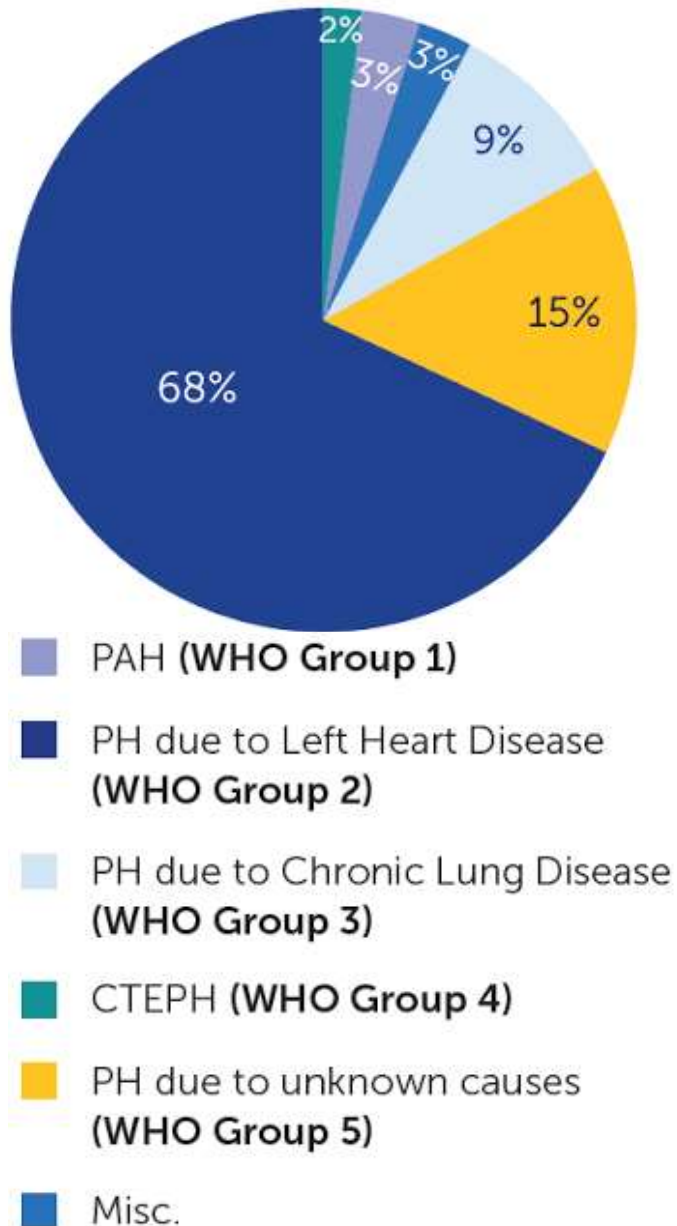
- **PAH:** Increased BP within pulmonary artery
- **Symptoms:** Shortness of breath, dizziness, chest pain, edema (leg, ankle, abdomen), cyanosis, fast heartbeat
- **Long term:** Vascular proliferation & remodeling
-> progressive increase in pulmonary vascular resistance -> right ventricle failure

Hemodynamic definitions of PAH	Pulmonary pressure
Mean PAP at rest	> 25 mmHg
Mean PAP with exercise	> 30 mmHg
Pulmonary capillary wedge pressure (PCWP or PWP)	< 15 mmHg
PAP = Pulmonary arterial pressure, PAP & PWP: assessed by right heart catheterization	



WHO Classification System of PH

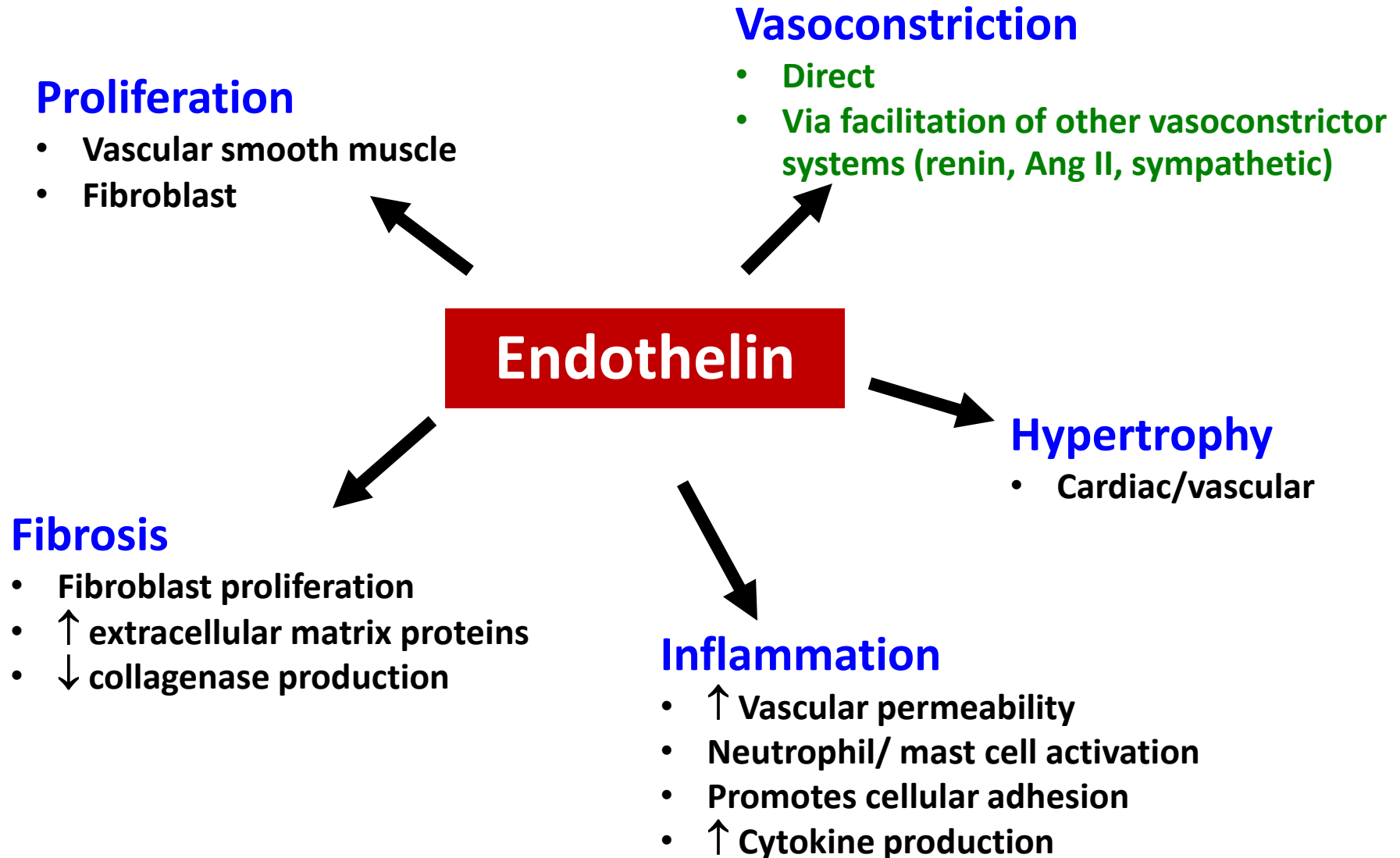
NYHA Functional Class of PAH



New York Heart Association Functional Classification

Class I	Patients that don't experience any symptoms with ordinary physical activity
Class II	Patients with symptoms while performing regular activities and slight limitation of activity
Class III	Patients that experience symptoms with less than normal activity, and marked limitation of activity
Class IV	Patients that suffer the symptoms of the disease with any activity or even at rest

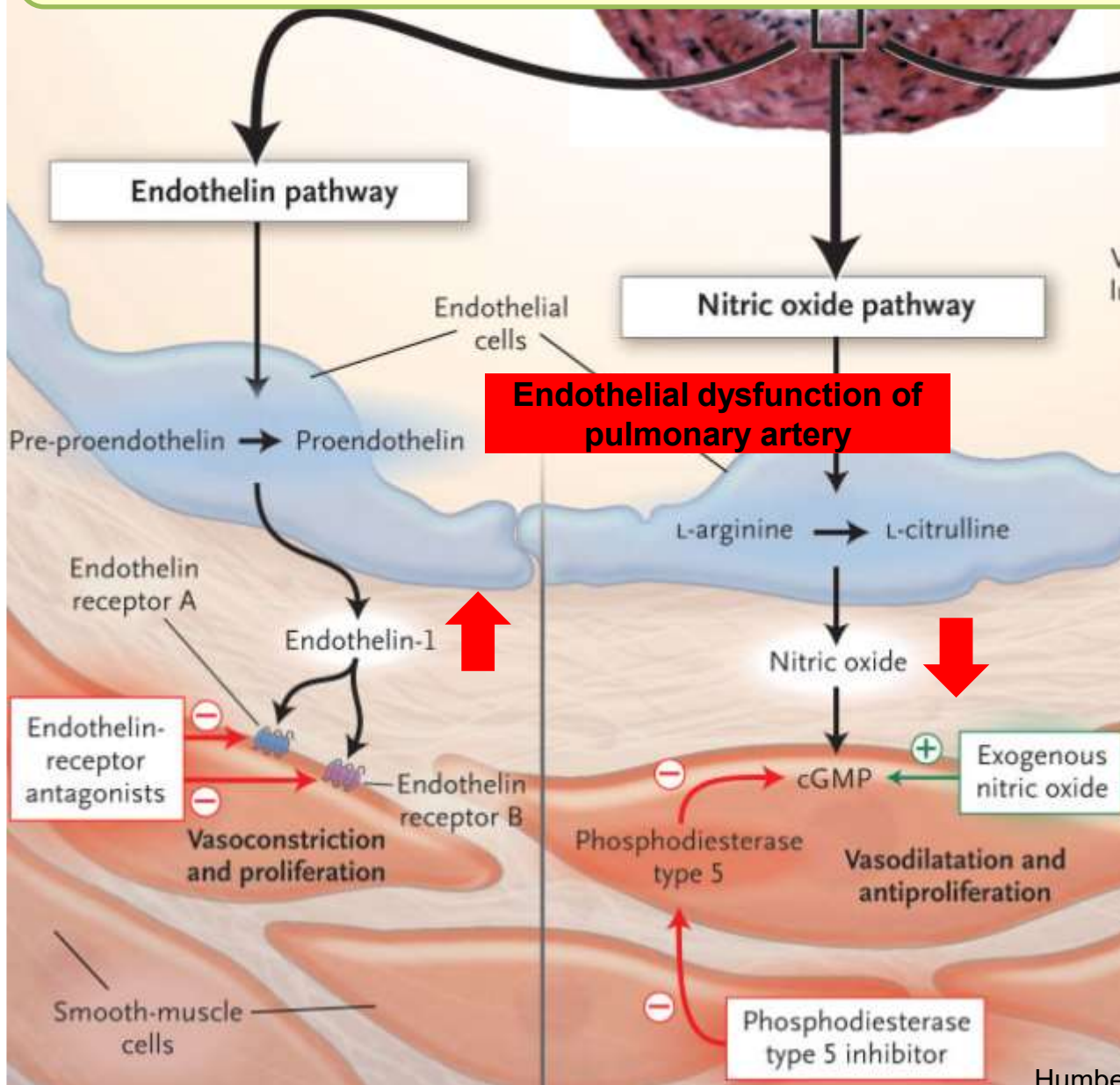
Endothelin is a Key Pathogenic Mediator



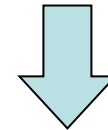
Endothelin-1 (ET-1) vs PAH

- Endothelin: ET-1, ET-2 and ET-3
 - ET-1 is produced mainly by the endothelial cell
- Endothelin receptors: ET_A & ET_B receptor
 - Vasoconstriction, found in vascular smooth muscle tissue
- Patients with PAH: Endothelial cell dysfunction → ↑↑↑ level of ET-1
 - Vasoconstriction
 - Inflammation
 - Fibrosis (tissue and blood vessels)
 - Hypertrophy & proliferation of cells (blood vessel narrow and occlusion)

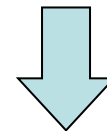
ERAs: Pharmacological effects



Endothelin receptor antagonists (ERAs)



↓ Vasoconstriction
↓ Inflammation
↓ Cell Proliferation
↓ Fibrosis



Treatment of PAH

ERAs: Pharmacokinetics

	Ambrisentan (Volibris®)	Bosentan (Tracleer®)	Macitentan (Opsumit®)
Selectivity ET _A :ET _B	200:1 Selective ET _A receptor	20:1 Dual ET _A & ET _B receptor	50:1 Dual ET _A & ET _B receptor
Absorption & Distribution			
Bioavailability	~ 90%	~ 50%	~ 74%
Food effect on absorption	No	No	No
t _{max} (h)	1.7 – 3.3	3 – 5	8
Albumin binding (%)	99	> 98	99
Metabolism & Excretion			
Elimination half-life (h)	15	5.4	16
Metabolism	Hepatic (3A4, 2C19, glucuronidation)	Hepatic (3A4, 2C9)	Hepatic (3A4)
Excretion	Biliary system	Biliary system	Kidney

ERAs: Therapeutic uses

	Ambrisentan (Volibris®)	Bosentan (Tracleer®)	Macitentan (Opsumit®)
Indication	PAH (WHO Group 1) with NYHA Functional Class II-IV symptoms to improve exercise ability and clinical symptoms		
Dosage	5-10 mg OD	62.5 mg BID for 4 wk, then increased to 125 mg BID	10 mg OD
Adverse effects	Hepatotoxicity, anemia, peripheral edema		
Drug Interactions	CYP3A4 inhibitor	CYP2C9 & 3A4 substrate/inducer/inhibitor	CYP3A4 substrate/inducer/inhibitor
Hepatic impairment	Not recommended	Not recommended	Not recommended
Renal impairment	No adjustment	No adjustment	No adjustment
Dosage form	Tab 5, 10 mg	Tab 62.5, 125 mg	Tab 10 mg
บ้ำญชียา	NED	NED	NED

PDE-5 inhibitors for PAH

	Sildenafil (Revatio®) (FDA approval, 2005)	Tadalafil (Adcirca®) (FDA approval, 2009)
Indication	PAH (WHO Group1) with NYHA Functional Class II-III symptoms to improve exercise ability and clinical symptoms	
Dosage	Oral: 20 mg TID, IV: 10 mg TID	Oral: 40 mg OD
Contraindications	Use with organic nitrates	Use with organic nitrates
Drug Interactions	CYP3A4 inducer / inhibitor	CYP2C9 & 3A4 inducer / inhibitor
Hepatic impairment	No dose adjustment	Mild/moderate: Start 20 mg Severe: Avoid
Renal impairment	No dose adjustment	Severe: Avoid
Dosage form	Tab 20 mg, IV 10 mg/12.5 mL	Tab 20 mg
บ้ญชียา	๑	NED

บัญชียาหลักแห่งชาติ พ.ศ. 2565

Sildenafil (as citrate)

tab

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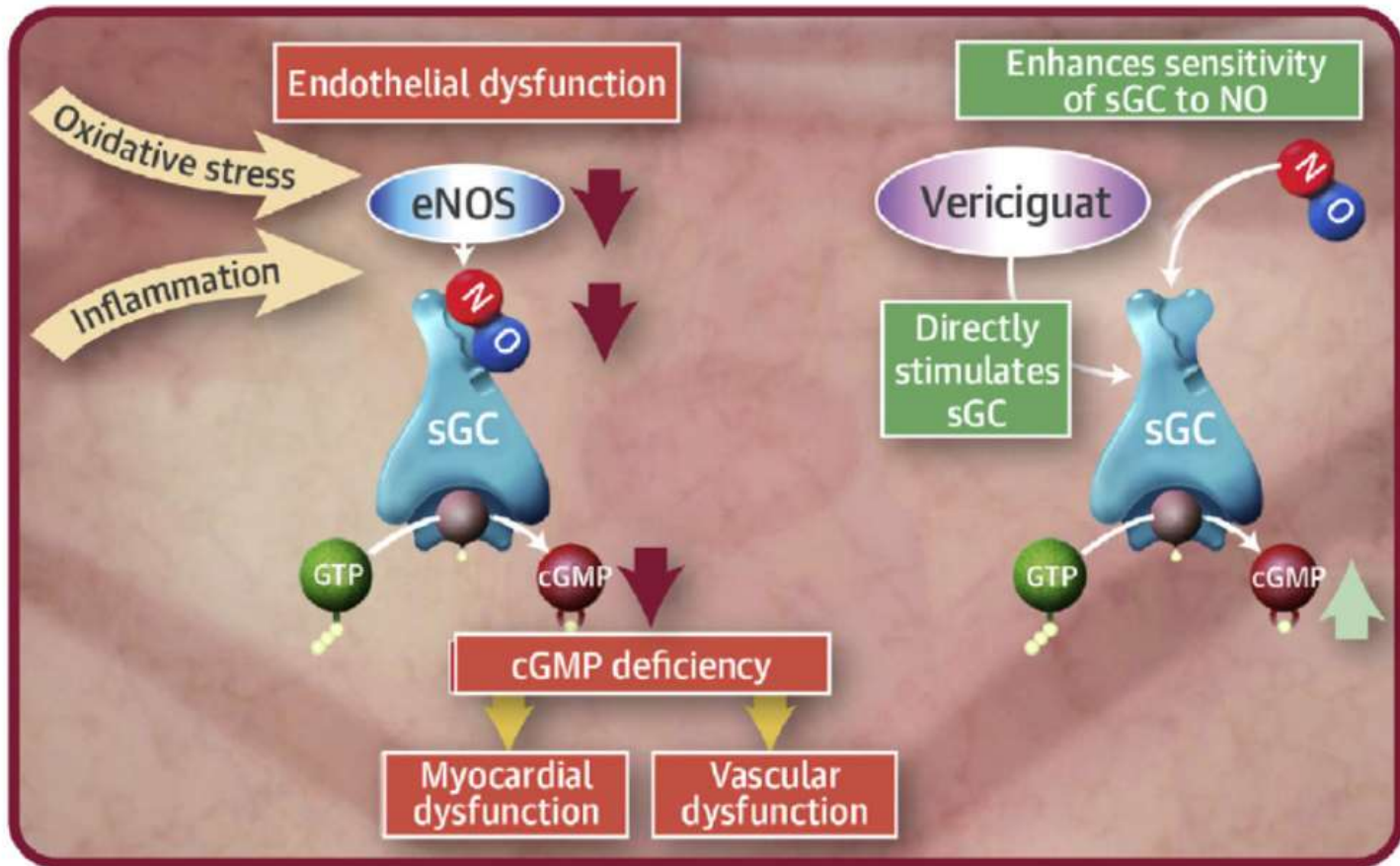
เงื่อนไข

1. ใช้สำหรับผู้ป่วยภาวะ pulmonary arterial hypertension (PAH) ที่เกิดจากโรคหัวใจแต่กำเนิด (CHD) ชนิด systemic-to-pulmonary shunt หรือโรค idiopathic pulmonary arterial hypertension (IPAH) หรือ PAH associated with connective tissue disease (CNTD) และ
2. อยู่ใน WHO functional classification of PAH $\geq$ II และ
3. ได้รับการตรวจวินิจฉัยตามขั้นตอนวิธีที่ปรากฏในแนวทางเวชปฏิบัติ
4. แนะนำให้หยุดยาเมื่อผลการประเมินทุก 3 เดือนมีอาการทรุดลงอย่างต่อเนื่องอย่างน้อย 2 รอบการประเมิน เกณฑ์อาการทรุดลงหมายถึงการตรวจพบอย่างน้อย 2 ข้อต่อไปนี้คือ
 - 4.1 ตรวจร่างกายมีอาการแสดงของ progressive right heart failure
 - 4.2 WHO functional classification เพิ่มขึ้นกว่าเดิม
 - 4.3 6MWT ลดลงกว่าเดิม 25%

Soluble guanylyl cyclase (sGC) stimulators

- **Rio**ciguat
- **Veri**ciguat

sGC stimulators: Pharmacological effects



Direct stimulation of soluble guanylate cyclase (sGC) -> ↑ formation of cGMP

sGC stimulators: Therapeutic uses

	Riociguat (Adempas®), 2013	Vericiguat (Verquvo®), 2021
Indication	Pulmonary arterial hypertension (PAH)	Reduce risk of CV death and hospitalization in patients with symptomatic chronic heart failure (EF< 45%)
Dosage	1 mg TID, titrate up every 2 wk, to 2.5 mg TID	2.5 mg OD, titrate 2x every 2 wk, to 10 mg OD
Pharmacokinetics	F 94%, T _{1/2} 5-10 hr, high protein binding Metabolism: CYP1A1, 3A4, 2C8	F 93% (with food), T _{1/2} 30 hr, high protein binding Metabolism: Glucuronidation
Adverse effects	Hypotension, ↑ liver enzymes, N/V, flushing	Hypotension, anemia
Drug Interactions	Nitrates, PDE-5 inhibitor	Nitrates, PDE-5 inhibitor
บ้ญพียา	NED	NED

Neprilysin inhibitors

- **Sacubutril**

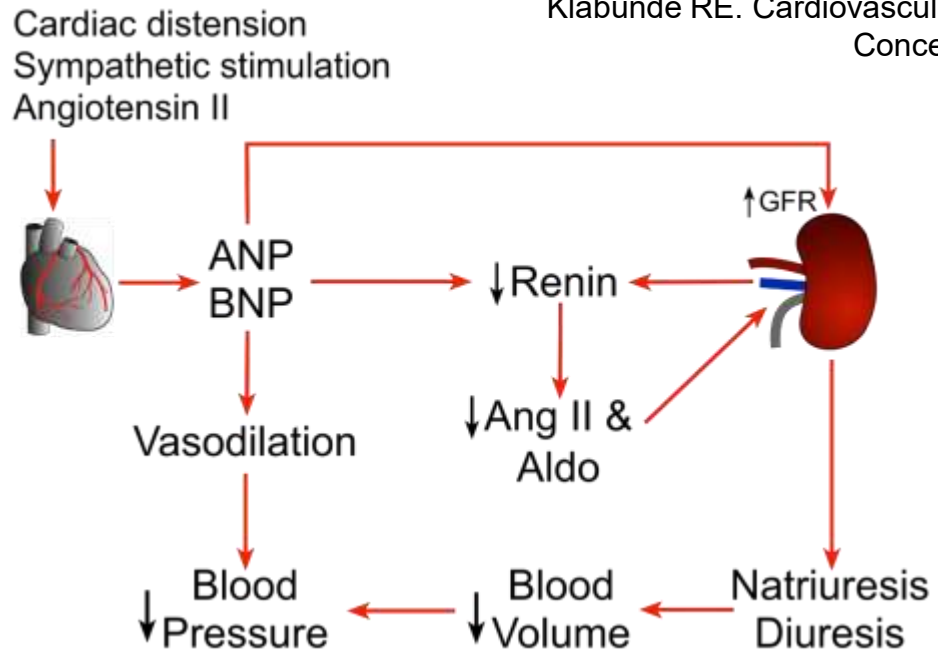
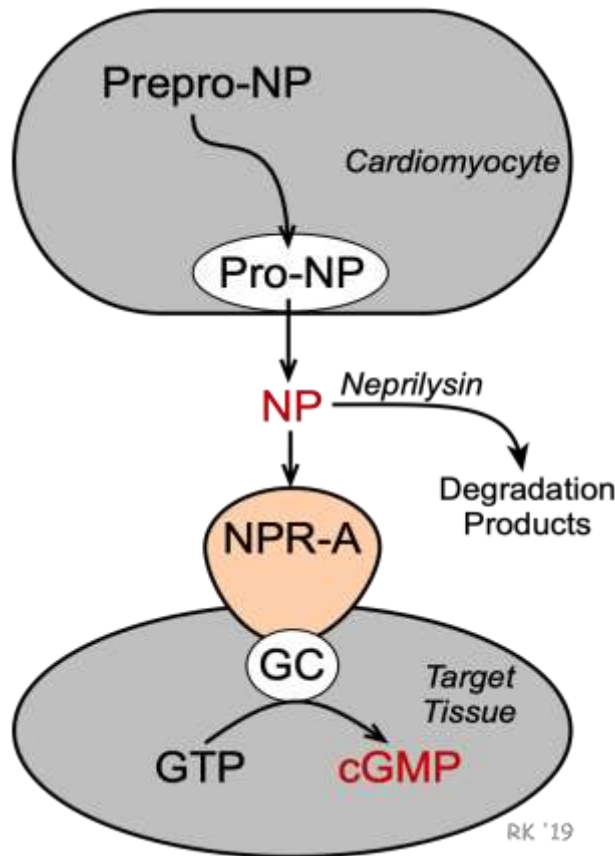
Natriuretic peptides (NPs)

- **NPs:** release from heart in response to pressure and volume overload
- **3 types of NP:** Atrial NP (ANP), B-type NP (BNP), C-type NP (CNP)

NPs	ANP	BNP	CNP
Location	Atria	Ventricle	Vascular endothelial cells
Releasing trigger	Atrial distension	↑ Ventricular volume	↑ Sheer stress
Receptor	NPR-A	NPR-A	NPR-B
Clearance of NP	Clearance via NPR-C via receptor-mediated internalization and degradation		
Enzymatic degradation	Degraded by neprilysin (NEP; neutral endopeptidase)		

Natriuretic peptides (NPs)

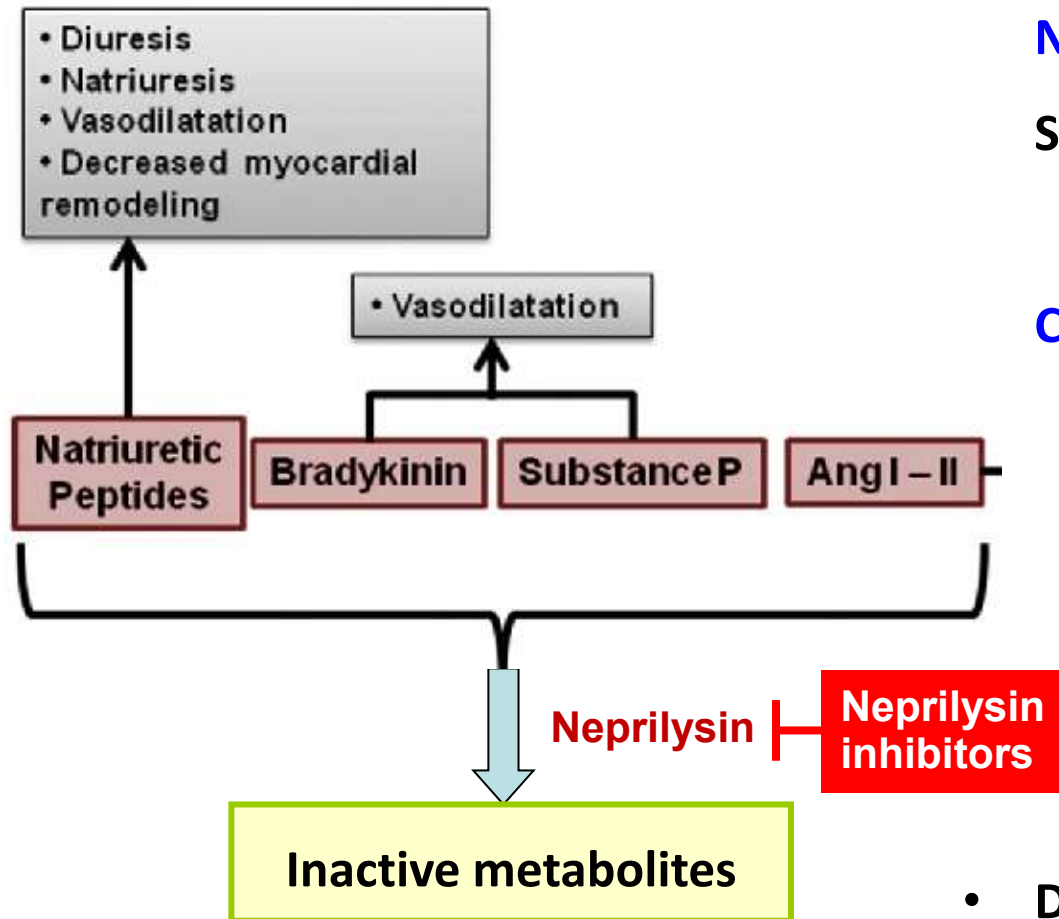
Klabunde RE. Cardiovascular Physiology
Concepts 2nd, 2012



Biological effects of NPs

ANP	BNP	CNP
▪ Natriuresis & diuresis ▪ Vasodilation ▪ RAAS & SNS suppress ▪ ↑ RBF & GFR ▪ Antihypertrophic ▪ Antifibrotic ▪ Anti-inflammatory	▪ Natriuresis & diuresis ▪ Vasodilation ▪ RAAS & SNS suppress ▪ ↑ RBF & GFR ▪ ↑ myocardial relaxation ▪ Antifibrotic	▪ Vasodilation ▪ Antihypertrophic ▪ Antifibrotic ▪ Anti-inflammatory ▪ Antithrombotic

Neprilysin inhibitors



Neprilysin inhibitors:

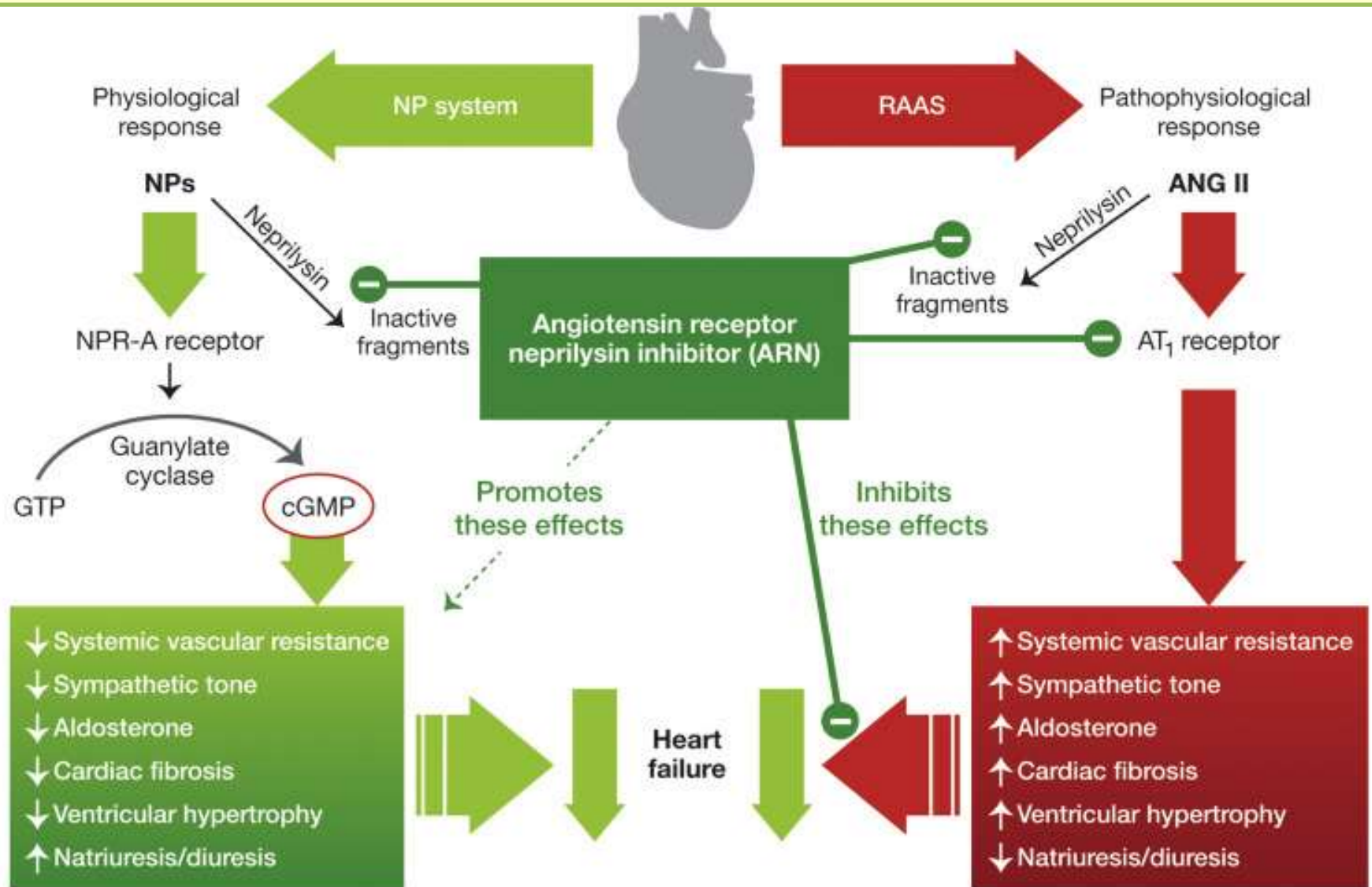
Sacubitril, Candoxatril

Candoxatril:

- Study in hypertension: $\uparrow$ ANP level, but $\leftrightarrow$ BP, $\uparrow$ Ang II level
- Using neprilysin inhibitor alone is not effective

- Drug development with **dual action**:
Neprilysin inhibitor +
RASS blockers (ACEIs or ARBs)

Angiotensin receptor - Neprilysin inhibitor (ARNI)



ARNI: Therapeutic uses

	Sacubitril / Valsartan (Entresto®) (FDA approval 2015)
Combination	Neprilysin inhibitor (Sacubitril) + Angiotensin II receptor blocker (ARB) (Valsartan)
Indication	- Heart failure (NYHA Class II-IV) and reduced ejection fraction - Heart failure with systemic left ventricular systolic dysfunction in pediatric (≥ 1 year, 2019)
Dosage	Start: 49/51 mg BID, then double to target maintenance dose of 97/103 mg BID
Adverse effects	Hyperkalemia, kidney dysfunction, hypotension Pregnancy Cat. X
Severe renal/hepatic impairment	Severe renal imp: 50% reduction Severe liver imp: Not recommended
Dosage form	Tab 24/26, 49/51, 97/103 mg
บัญชียา	NED